

Dylan Kawalec

AI Solutions Architect (Confidential Systems)

Forward Deployed Engineering | Applied Research | Protocol Integrations

Building the trust layer for private intelligent systems

SF Bay Area | [linkedin.com/in/dylankawalec](https://www.linkedin.com/in/dylankawalec) | github.com/DylanCkawalec | jokuh.com | dylankawalec@gmail.com

Professional Summary

AI Solutions Architect for confidential systems. I build production private AI runtimes, TEE-backed deployment paths, and protocol integrations, then use research artifacts to validate and formalize the underlying architecture. Recent work includes Phala dstack deployments for 24+ partner teams, Rust SGX tooling for Osmosis on Azure, and zero-trust verifier prototypes in Python/C++. Most roles were contract or founding engagements across early-stage infrastructure teams.

Core Competencies

Technologies: Python, Rust, C/C++, Go, TypeScript, Solidity, PyTorch, FastAPI, Docker, GitHub Actions, Postgres, Redis.

AI Systems Built: local-first inference pipelines (Ollama, llama.cpp, vLLM; Llama/Mixtral/Qwen); TEE-backed agents; verifier prototypes; Rust deployment CLIs.

Confidential Compute & Cloud: Intel TDX/SGX, Phala Cloud/dstack, Fortanix, Azure SGX VMs, AWS, Cloudflare, remote attestation, enclave TLS.

Protocol Specializations: protocol integrations, partner engineering, ZK authentication, threshold signing, threat modeling, TLA+/PlusCal specs.

Production Systems & Research Artifacts

- **Production:** Built jokuh.com for designing and deploying local-first AI systems with agent orchestration, architecture planning, and zero-trust execution patterns.
- **Production:** Shipped dstack integrations on Phala, including [Warp Engine](#) for attested TDX private-agent execution and an Akave gateway for confidential o3 inference paths.
- **Production / open source:** Led SGX decentralized front-end research and built github.com/osmosis-labs/defe, a Rust CLI for Fortanix SGX provisioning on Azure.
- **Research:** Published github.com/DylanCkawalec/DI and ENI6MA/Cosmic verifier prototypes to demonstrate ERC-8004 agent identity, private AI authentication, and zero-trust proof boundaries.

Experience

Head of AI, Decentralized Confidential Compute *Jokuh (Independent R&D)*

Dec 2025 – Present

- Building and operating production confidential AI systems for private inference and multi-agent orchestration, including Mermate, Synth, and Jokuh workflows across local and cloud environments.

Head of Developer Relations & Solutions Architect (Protocol Integrations) *Phala Network*

May – Dec 2025

- Led confidential AI rollout on Intel TDX via dstack; authored Docker reference deployments and integration docs that moved 24+ partner teams from evaluation into production deployments of confidential AI systems.
- Built protocol integrations across EigenLayer, Redpill.ai, Rift.trade, Akave, and TEE agent teams; grew dstack visibility 34→400+ GitHub stars and co-authored Trustless Cloud Agents (CCC 17/17 vote).

Founding Engineer / Principal Architect *Cosmic Inc. | ENI6MA Research*

Summer 2024 – Jan 2026

- Designed zero-trust architecture (TEE-only execution + cryptographic proof boundaries); built Python/C++ verifiers, local private inference pipelines, protocol specs, threat models, and deterministic replay tests.
- Delivered provisional patent architecture supporting a \$168K friends-and-family raise; published public research excerpts and proof previews via eni6ma.gitbook.io/eni6ma.

Head of Developer Relations (Protocol Integrations) *GOAT Network (BTC L2)*

Jul 2024 – Jan 2025

- Delivered Founders Club MVP and onboarded 100+ founders; authored zkMIPS/EVM docs, ran BTC L2 workshops, and supported ZKM-led Bitcoin L2 integration research.

R&D Engineer *Osmosis Foundation*

May – Jul 2024

- Shipped production Rust CLI ([defe](#)) for Fortanix SGX on Azure; automated enclave TLS, key lifecycle, and threshold signing for decentralized front-ends.

CTO & Principal Engineer *Invo Technologies*

Apr 2023 – Feb 2024

- Built the Invo MVP on AWS; completed patent architecture, led github.com/Invo-Technologies/poems, and won Aleo Best Tech 2023 (\$50K grant).

Earlier: BNB Chain Labs (early U.S. technology evangelist; partner integrations with Polyhedra, StorJ, QuickNode, Bware Labs) • Findora/Discreet Labs (Protocol Integrations; 80+ validators; adoption during major funding expansion) • Harmony (Developer Captain/DAO Governor; \$100K DAO funding; 700+ devs) • Tesla (Model 3 Technician) • Zygo (IBF Technician; precision optics, JWST metrology).

Recognition, Credentials & Publications

Awards: ETHGlobal SF 2024 (Ledger 1st, Polygon 1st), Aleo Best Tech 2023, VeChain×EasyA PowerUp (2nd), ETHDenver 2024 Hacker Deathmatch.

Credentials: RareSkills Research Fellow; ZKU Developer; B.S. Computer Science, WGU (2019–2021); A.A.S. Computer Programming, MiraCosta (2015–2019).

Selected links: phalanetwork.mintlify.app/docs | docs.goat.network | eni6ma.gitbook.io/eni6ma | github.com/DylanCkawalec/cosmic-hack | blackbox-podcast.com